Scalar Equation of a Line

Another way to form the equation of a line is to use a vector that is $\perp$ to the line rather one that is parallel to the line.

This is called the "normal vector".

Scalar / cartesian Equation of a Line is $Ax + By + C = 0$; where vector $[A, B]$ is a normal to the line.

Example 1: Find a normal to the line.

$$\vec{r} = [1, 3] + t[5, 2]$$

Solution: $\vec{d} = [d_1, d_2]$

normal vector : $[-d_2, d_1]$

$[-2, 5]$

Example 2: Find the scalar equation of line with normal $[1,3]$ which passes through point $(-4,2)$.

<u>Solution:</u>

$$Ax + By + C = 0$$

$$(1)(-4) + (3)(2) + C = 0$$

$$2 + C = 0$$

$$C = -2$$

$$\therefore \quad x + 3y - 2 = 0 \left.\right\} \text{Scalar Equation.}$$

note: Given normal vector $[A,B]$ the slope of the line is $m = -\frac{A}{B}$

Intersection of Lines

4 possible scenarios for the relationship between lines in 3 space.

1. parallel and identical (coincident)
2. parallel and non-identical (distinct no solution)
3. non-parallel and intersecting (1 solution)
4. non-parallel and non-intersecting ("skew lines" no solutions)

Example 1: Find the intersection of

l_1: $x = 3 + 2k$ $\quad$ l_2: $x = 5 - 4w$

$y = 4 - 3k$ $\quad$ $y = 2 + 6w$

$z = 5 + k$ $\quad$ $z = 4 - 2w$

Step 1: $\vec{d_1} = [2, -3, 1]$

$\vec{d_2} = [-4, 6, -2]$

$\therefore$ $\vec{d_2} = -2\vec{d_1}$ $\Rightarrow$ l_1 and l_2 are parallel.

Step 2: check if they are coincident

$3 = 5 - 4w$	$4 = 2 + 6w$	$5 = 4 - 2w$
$w = \frac{1}{2}$	$w = \frac{1}{3}$	$w = \frac{-1}{2}$

w is <u>not</u> the same

$\Rightarrow$ not the same line.

$\therefore$ l_1 and l_2 are parallel <u>and</u> distinct.

Example 2: Find point of intersection

l_1: $x = 3 + 2k$ $\quad$ l_2: $x = -7 + 6t$

$y = 4 - 3k$ $\quad$ $y = 5 + 5t$

$z = 5 + k$ $\quad$ $z = -1 + 4t$

Step 1: $\vec{d_1} = [2, -3, 1]$; $\vec{d_2} = [6, 5, 4]$

since $\vec{d_1} \neq c\vec{d_2} \Rightarrow$ not parallel.

scalar

At pt. of intersection

$$3 + 2k = -7 + 6t \Rightarrow 2k - 6t = -10 \quad (1)$$

$$4 - 3k = 5 + 5t \Rightarrow -3k - 5t = 1 \quad (2)$$

$$5 + k = -1 + 4t \Rightarrow k - 4t = -6 \quad (3)$$

(1) - 2 x (3)

$$\begin{aligned} 2k - 6t &= -10 \\ -(2k - 8t &= -12) \\ \hline 2t &= 2 \\ \boxed{t = 1} \end{aligned}$$

Sub. t=1 into (3) to find k

$$k - 4(1) = -6$$

$$\boxed{k = -2}$$

check using (2)

L.S.	R.S
$-3(-2) - 5(1)$ $= 1$	1

$\Rightarrow$ L.S. = R.S,

$\therefore$ pt of intersection is $(-1, 10, 3)$

Example 3: Find point of intersection, if any.

$l_1: [-2, 1, 0] + s[1, 3, 7]$

$l_2: \dfrac{x-3}{5} = \dfrac{y+3}{-4} = \dfrac{z-4}{2}$

Step 1: $\vec{d_1} = [1, 3, 7]$; $\vec{d_2} = [5, -4, 2]$

$\therefore$ since $\vec{d_1} \neq k\vec{d_2} \Rightarrow$ not parallel

At the pt of intersection

l_1: $x = -2 + s$ $\quad$ l_2: $x = 3 + 5t$

$y = 1 + 3s$ $\quad$ $y = -3 - 4t$

$z = 7s$ $\quad$ $z = 4 + 2t$

$-2 + s = 3 + 5t \Rightarrow s - 5t = 5$ ①

$1 + 3s = -3 - 4t \Rightarrow 3s + 4t = -4$ ②

$7s = 4 + 2t \Rightarrow 7s - 2t = 4$ ③

$3 \times$ ① $-$ ②

$3s - 15t = 15$

$-(3s + 4t = -4)$

$-19t = 19$

$\boxed{t = -1}$

sub $t = -1$ into ①

$s - 5(-1) = 5$

$\boxed{s = 0}$

check using ③

L.S.	R.S
$7(0) - 2(-1)$ $= 2$	4

$\therefore$ No solution

$\therefore$ no int. point

SKEW LINES!